

Delayed diagnosis of invasive macroprolactinoma in a 16-year-old boy



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Background

Prolactinoma is the most common pituitary tumor, with heterogeneous clinical presentations according to sex, age of onset, tumor size, and prolactin levels. These tumors are rare in the pediatric and adolescent populations, with an estimated incidence of 0.1 per million and accounting for less than 2% of all intracranial tumors in children.

Case presentation

We report the case of a 16-year-old male adolescent referred to the department of endocrinology for management of a pituitary macroadenoma discovered after worsening chronic posterior and fronto-orbital headaches that were refractory to analgesics and had evolved over 6 years. He was the first child of non-consanguineous parents and had no significant medical conditions other than a family history of hypothyroidism.

- Clinical examination revealed reduced visual acuity, a scotoma-type visual field defect, memory and attention disturbances, dizziness, and delayed puberty (Tanner stage G4P4). Bone age was estimated at 15 years.
- Pituitary hormone testing showed marked hyperprolactinemia at 4786 ng/mL, secondary hypothyroidism (low FT4 with an inappropriately normal TSH), hypogonadotropic hypogonadism (low testosterone with low-normal LH and FSH), secondary adrenal insufficiency (low 8 a.m. cortisol with an inappropriately normal ACTH), and a low IGF-1 level suggestive of growth hormone deficiency (Table 1).
- MRI pituitary revealed a 25×29×17 mm macroadenoma, partially cystic, with suprasellar extension, invading the left cavernous sinus, classified as Knosp grade III, and displacing the optic chiasm, without signs of intracranial hypertension.

Table 01: Pituitary Hormone Profile

Hormone	Unit	Result	Reference Interval
Prolactin	ng/mL	4786	4.04 – 15.2
TSH	μIU/mL	2.29	0.27 – 4.2
FT4	pg/mL	4.87	11 – 18
IGF-1	ng/mL	96.4	125 – 503
8 a.m. Cortisol	nmol/L	109	171 – 536
ACTH	pg/mL	20.8	< 46
Testosterone	ng/dL	60.8	129 – 767
FSH	mIU/L	3.89	1.5 – 12.9
LH	mIU/L	2.28	1.7 – 8.6

References

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Discussion

This case illustrates a severe presentation of a pituitary macroadenoma in the peripubertal period, characterized by a large tumor burden, mass effect on adjacent structures, and extensive hypopituitarism affecting gonadotrophic, corticotrophic, thyrotrophic and growth hormone axes (Figure 1).

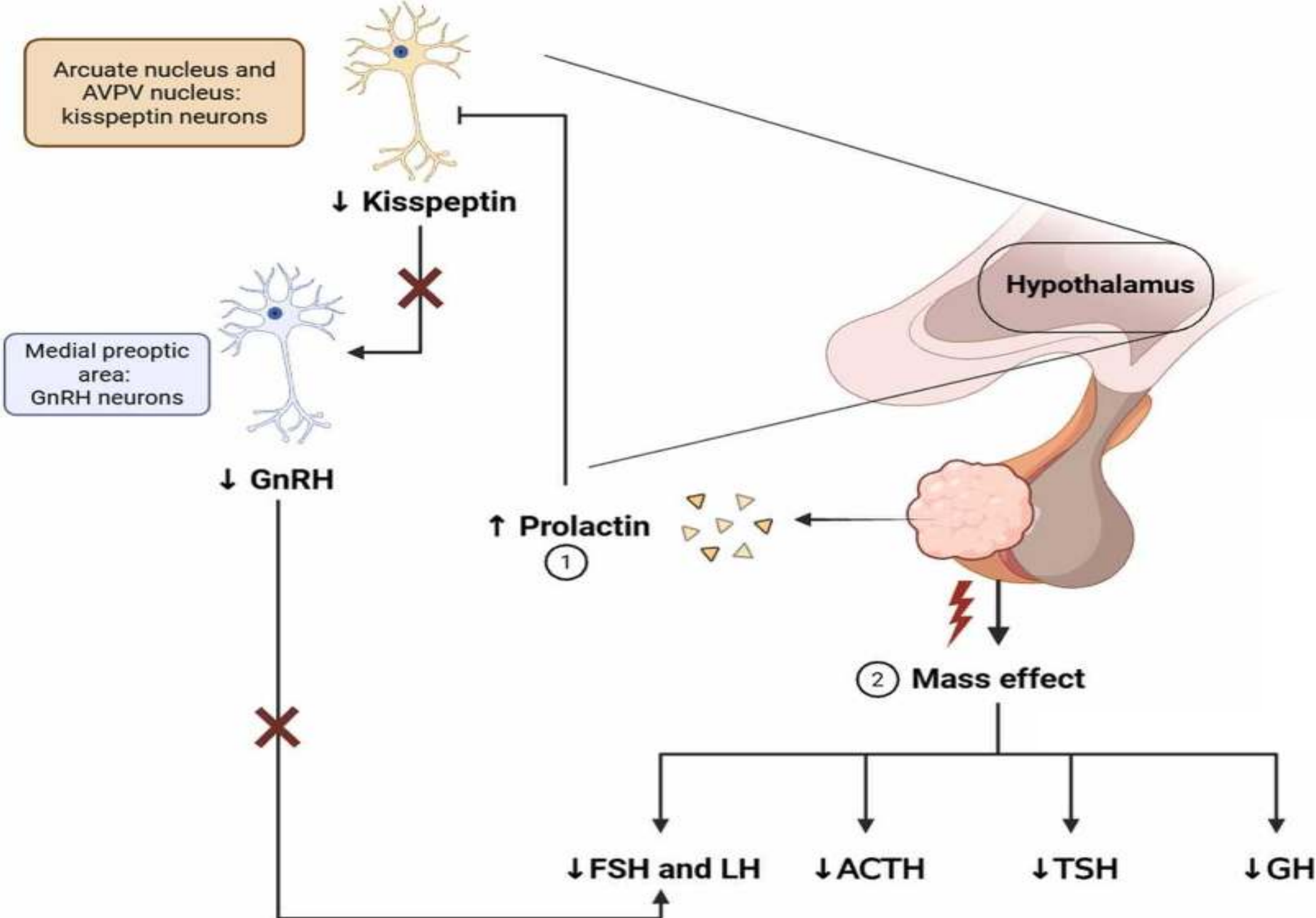


Figure 1. Pathophysiology of hypopituitarism development in prolactinomas
Adapted from Campana et al. (2026)

- The patient received the dopamine agonist (cabergoline) at a dose of 0.5 mg/week, which was increased gradually to 2.5 mg/week.
- Three months later, there was remarkable improvement of the hyperprolactinemia, with prolactin concentrations decreasing by more than 99% (38 ng/mL).
- There was also improvement in visual symptoms and headaches, reflecting the efficacy of treatment on both tumor size and hormonal secretion (Figure 2).
- Hormone replacement therapy with levothyroxine and hydrocortisone was initiated, with ongoing endocrine and radiological follow-up.
- The presence of a family history of endocrine disease warrants family screening for multiple endocrine neoplasia type 1 (MEN1) and genetic testing.

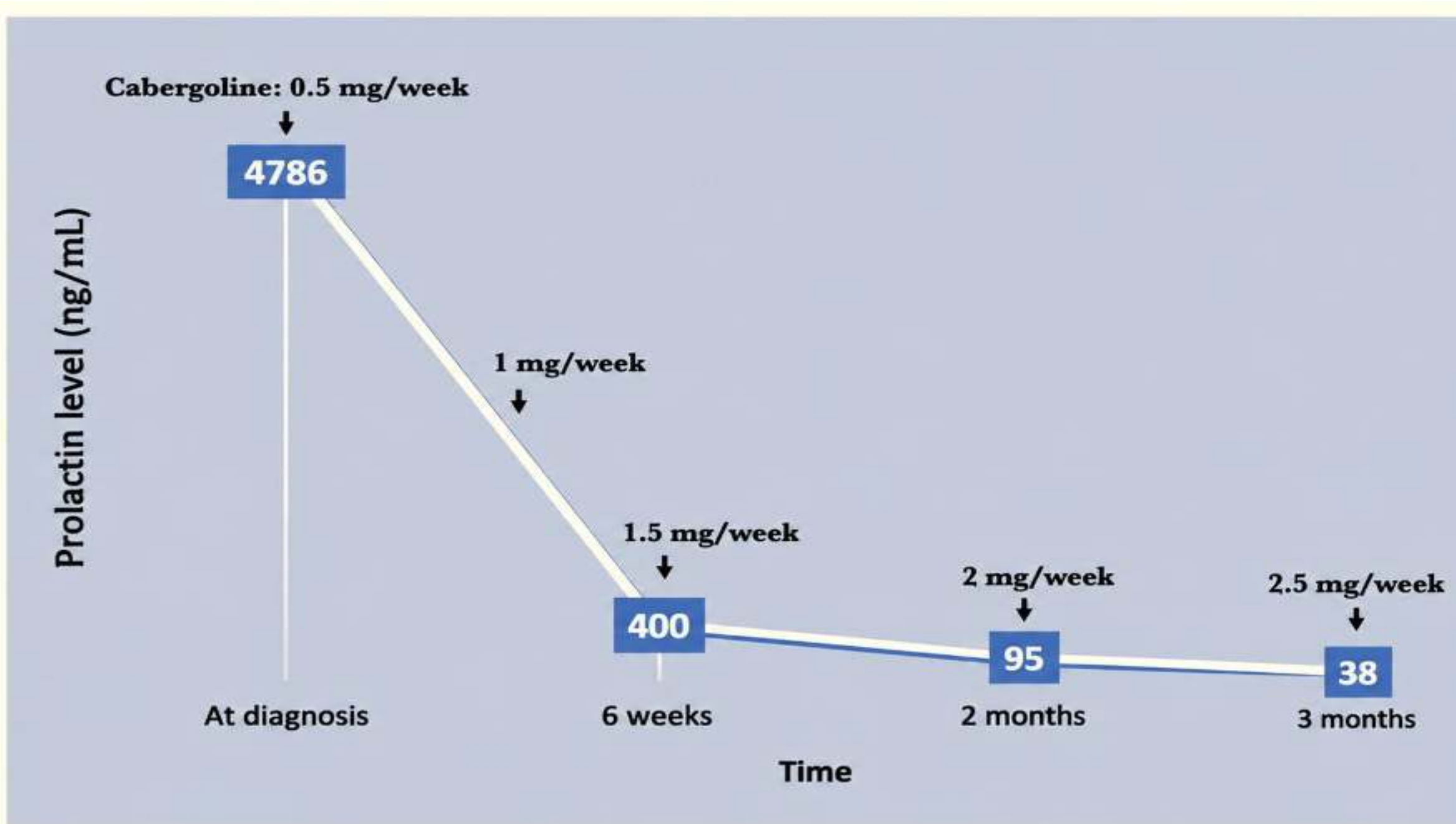


Figure 2. Changes in prolactin concentrations during treatment

Conclusion

- Pediatric prolactinomas are uncommon and may present with multiple pituitary hormone deficiencies and compressive symptoms.
- Early diagnosis and treatment are essential to achieve favorable clinical and biochemical outcomes.